

Security+ 4.4 · Security Tools

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.4** —
Security Tools (14:06) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: SCAP (Security Content Automation Protocol) is a standard for automating compliance checks (e.g., NIST 800-53, CIS). Other security tools like SIEM, SOAR, vulnerability scanners, FIM, and EDR each cover specific stages of security operations.

High-Yield Core Concepts Tested on Exam

- SIEM detects; SOAR acts; EDR isolates; FIM checks file integrity
- SCAP automates security compliance checks
- SOAR enables automated response and orchestration
- EDR provides endpoint detection and response capabilities
- FIM monitors changes to critical files
- Vulnerability scanners identify weaknesses in systems

Exam Tips & Classic Scenarios

- If the question mentions 'automates security compliance checks', the answer is SCAP or OpenSCAP
- Know the difference: SIEM detects, SOAR responds, EDR isolates, FIM monitors files

Typical Exam Scenario: A company needs to automate compliance checking against NIST 800-53. They implement SCAP to perform automated security compliance checks.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.4 — Security Tools)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **SIEM:** Security Information and Event Management: Centralized platform performing real-time ingestion, normalization, correlation, and alerting across multi-source security telemetry.
- **SOAR:** Security Orchestration, Automation, and Response: Automation platform orchestrating SOC workflows and incident containment through standardized playbooks.
- **EDR:** Endpoint Detection and Response: Agent installed on endpoints providing behavioral telemetry, threat detection, investigation capabilities, and live remote isolation.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. Platform that AUTOMATES SOC response (e.g., isolates the machine by itself upon seeing the alert) is:

- A) SIEM
- B) SOAR
- C) EDR
- D) VPN

2. Which solution AUTOMATES and orchestrates incident response, integrating with SIEM through playbooks?

- A)** Traditional antivirus
- B)** SOAR (Security Orchestration, Automation and Response)
- C)** Incremental backup
- D)** Edge firewall

3. Corporate scenario (Security Tools): During a security review and compliance audit for the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'SIEM detects; SOAR acts; EDR isolates; FIM checks files'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: SIEM detects; SOAR acts; EDR isolates; FIM checks files...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

4. Corporate scenario (Security Tools): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'SCAP: automates compliance check.' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: SCAP: automates compliance check...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

5. Corporate scenario (Security Tools): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'automates security compliance checks' = SCAP/OpenSCAP'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'automates security compliance checks' = SCAP/OpenSCAP...

6. Corporate scenario (Security Tools): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'company automates NIST compliance check' = SCAP. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'company automates NIST compliance check' = SCAP...

7. Corporate scenario (Security Tools): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'SIEM detects; SOAR ACTS; EDR isolates; FIM checks files'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: SIEM detects; SOAR ACTS; EDR isolates; FIM checks files.
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Security Tools): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'SCAP: automates compliance check'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: SCAP: automates compliance check...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Security Tools): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'automates security compliance checks' = SCAP/OpenSCAP. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'automates security compliance checks' = SCAP/OpenSCAP...

10. Corporate scenario (Security Tools): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'company automates NIST compliance check'=SCAP'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'company automates NIST compliance check'=SCAP...
- D)** Allow shared access among administrators with generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — SOAR.

✓ **Why it is correct:** SOAR (Security Orchestration, Automation and Response) is a platform that integrates security tools, orchestrates workflows, and automates incident response. In the context of CompTIA Security+ (SY0-701), SOAR is highlighted for its ability to execute automatic actions, such as isolating a compromised machine, blocking a malicious IP, or disabling a user account, without human intervention. The statement describes exactly a platform that automates SOC response, such as isolating the machine by itself upon seeing the alert, which is the primary function of SOAR.

✗ **Why the others are wrong:**

- **SIEM:** SIEM (Security Information and Event Management) is a solution that collects, correlates, and analyzes logs and security events in real time, generating alerts for SOC teams. Although essential for detection, SIEM traditionally does not execute automated response actions; it only notifies. Response automation is a SOAR capability, which can integrate with SIEM. Therefore, SIEM is not the platform that automates response as described.
- **EDR:** EDR (Endpoint Detection and Response) is a solution focused on endpoints (workstations, servers) that monitors suspicious activities, detects threats, and allows direct response on the device, such as isolating the machine. However, EDR acts primarily on the endpoint and does not orchestrate responses across the entire infrastructure. The statement speaks of a platform that automates SOC response broadly, which is the role of SOAR. EDR can be one of the tools integrated into SOAR, but it is not the SOC response automation platform.
- **VPN:** VPN (Virtual Private Network) is a technology that creates an encrypted tunnel for secure communication between networks or remote users. It has no relation to incident response automation or security orchestration. Therefore, it is completely inapplicable to the context of the statement.

2) B — SOAR (Security Orchestration, Automation and Response).

✓ **Why it is correct:** The correct answer is 'SOAR (Security Orchestration, Automation and Response)'. SOAR is a platform that integrates and orchestrates security tools, automates repetitive incident response tasks, and enables the creation of playbooks (automated workflows). It connects to SIEM (Security Information and Event Management) to receive alerts and, based on them, execute actions such as blocking an IP, isolating an endpoint, notifying analysts, or opening tickets, all in an automated manner. In CompTIA Security+, SOAR is presented as an evolution that increases SOC (Security Operations Center) efficiency, reducing response time and alert fatigue, while standardizing procedures.

✗ **Why the others are wrong:**

- **Traditional antivirus:** Traditional antivirus is a signature-based solution that detects and removes known malware on endpoints. It does not orchestrate incident responses, does not natively integrate with SIEM for playbook automation, and lacks orchestration capabilities across multiple security tools. Its scope is limited to host protection, and incident response is typically manual and localized. Therefore, it does not meet the requirement to automate and orchestrate incident response in an integrated manner.
- **Incremental backup:** Incremental backup is a data copying strategy that records only changes since the last backup (full or incremental). It is a data recovery and business continuity measure, not an automated incident response tool. Although essential for restoring systems after an attack, it does not orchestrate containment, eradication, or notification actions, nor does it integrate with SIEM through playbooks. Its function is purely information protection and recovery, not incident

response automation.

- *Edge firewall*: An edge firewall is a network security device that filters traffic based on access control rules (IP, port, protocol). It can block unwanted connections but does not automate incident response in an orchestrated manner, nor does it integrate with SIEM to execute complex playbooks. Traditional firewalls lack multi-tool orchestration, workflow automation, or alert enrichment capabilities. They are perimeter defense components but do not replace a SOAR platform.

3) A — Implement the corresponding solution: SIEM detects; SOAR acts; EDR isolates; FIM checks files....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Tools): SIEM detects; SOAR acts; EDR isolates; FIM checks files.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: This destroys individual accountability and non-repudiation. It makes it impossible to prove who made critical changes or malicious commands.

4) A — Implement the corresponding solution: SCAP: automates compliance check....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Tools): SCAP: automates compliance check. SCAP (Security Content Automation Protocol) is a suite of specifications that standardizes the format and nomenclature for expressing and manipulating security-related information, enabling automated vulnerability management, measurement, and policy compliance evaluation.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

5) D — Implement the corresponding solution: 'automates security compliance checks' = SCAP/OpenSCAP....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Tools): 'automates security compliance checks' = SCAP/OpenSCAP. SCAP (Security Content Automation Protocol) is a suite of specifications that standardizes the automation of vulnerability management, measurement, and policy compliance evaluation; OpenSCAP is an open-source implementation.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) D — Implement the corresponding solution: 'company automates NIST compliance check' = SCAP....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Tools): 'company automates NIST compliance check' = SCAP. The Security Content Automation Protocol (SCAP) is a suite of specifications that standardizes the format and nomenclature for expressing and manipulating security-related information. It enables automated vulnerability management, measurement, and policy compliance evaluation against standards like NIST guidelines.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) B — Implement the corresponding solution: SIEM detects; SOAR ACTS; EDR isolates; FIM checks files..

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Tools): SIEM detects; SOAR ACTS; EDR isolates; FIM checks files. This mnemonic summarizes the primary functions of key security tools: SIEM (Security Information and Event Management) aggregates and analyzes logs to detect threats; SOAR (Security Orchestration, Automation, and Response) automates incident response actions; EDR (Endpoint Detection and Response) monitors and isolates endpoints; FIM (File Integrity Monitoring) detects unauthorized changes to files. Security+ expects familiarity with these tools and their roles.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaches compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

8) A — Implement the corresponding solution: SCAP: automates compliance check....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Tools): SCAP: automates compliance check. SCAP (Security Content Automation Protocol) is a suite of specifications that standardizes the automation of vulnerability management, measurement, and policy compliance evaluation.

✗ **Why the others are wrong:**

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- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) D — Implement the corresponding solution: 'automates security compliance checks' = SCAP/OpenSCAP....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Tools): 'automates security compliance checks' = SCAP/OpenSCAP. SCAP (Security Content Automation Protocol) is a suite of specifications that standardizes the format and nomenclature for security automation, enabling automated vulnerability management, measurement, and policy compliance evaluation. OpenSCAP is an open-source implementation that uses SCAP to scan systems against security baselines (e.g., DISA STIGs, PCI DSS, ISO 27001) and generate compliance reports.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

10) C — Implement the corresponding solution: 'company automates NIST compliance check'=SCAP....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Tools): 'company automates NIST compliance check'=SCAP. SCAP (Security Content Automation Protocol) is a suite of specifications that enables automated vulnerability management, measurement, and policy compliance evaluation against standards like NIST.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks,

injections, or malware at runtime.

- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.